



CSE471: System Analysis and Design
Assignment on Functional Requirements

Proposed Project Title : Student Collaboration Hub

Group No: 9, CSE471 Lab Section: 3, Summer 2026		
SL	ID	Name
1	23101224	Rayhanul Hoque Rayhan
2	22299376	Fouzia Ahmed Reya
3	23301127	Sayeeefa Sayeed
4	23301297	Tuli Saha

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Project Overview

The “**Student Collaboration Hub**” is a comprehensive web-based platform which will be designed for helping the university students to collaborate efficiently throughout their academic journey. The system will support three distinct user roles: Students, Tutors, and Administrators. It will connect students by allowing them to create study groups, find project teammates, manage academic projects, share academic resources, communicate with peers and receive AI-powered recommendations. Also it will provide a marketplace for buying and selling books and students will be able to schedule tutoring sessions and meetings.

Tech Stack:

- Language: PHP , Javascript
- Framework: Laravel
- Styling: Bootstrap 5, CSS
- Database: MySQL
- Deployment Environment: XAMPP
- Version Control: Git & GitHub
- Deployment: InfinityFree
- External APIs:
 - **Google Maps API:** Display nearby study locations, libraries, and meeting locations. (Rayhan)
 - **Cloudinary API:** For uploading and managing profile pictures, notes, and documents. (Fouzia)
 - **Google Calendar API:** For meeting scheduling, event synchronization and managing project deadlines. (Sayeefa)
 - **Google Gemini API:** For generating AI-powered project ideas and recommending suitable teammates. (Tuli)

User Roles

1. **Student:** Students can register, update their profiles, search for teammates, create study groups, upload notes, buy or sell used books, communicate with teammates, manage projects, and participate in academic events.
2. **Tutor:** Tutors can create tutoring profiles, receive tutoring requests, manage tutoring sessions, upload learning materials, and communicate with students.

3. **Administrator:** Administrators manage users, verify tutor accounts, moderate uploaded resources, monitor reported content, manage announcements, and oversee the overall system.

Functional Requirements

SL	Common Workflows
1	Registration and Login System: Students and tutors can register using their university email address, personal information, and password. After registration, users can securely log in to access their personalized dashboard. Password recovery and email verification are also supported.
2	Admin Role & Access Management: Administrators access a dedicated dashboard where they can verify tutor registrations, monitor user activities, remove inappropriate content, manage reports, and maintain overall platform security. Different user roles receive different permissions based on their responsibilities.

Module 1	
Member	Feature Description
(Rayhan)	Student Profile & Skill Management: Students can create and update their profiles by adding personal information, department, semester, technical skills, interests, completed projects, portfolio links and preferred study locations using Google Maps.
(Fouzia)	Notes Sharing System: Students can upload, edit, organize, search, and download lecture notes and study materials. Files and images are stored using Cloudinary.

(sayeefa)	Project Task Management: Students can create projects, add tasks, assign responsibilities, set deadlines, and monitor task completion. Tasks can also be synchronized with Google Calendar.
(Tuli)	AI Project Idea Generator: Students can enter a project domain or keywords, and the system generates project ideas, suggested technologies, and implementation plans using the Google Gemini API.

Module 2	
Member	Feature Description
(Rayhan)	Study Group Management: Students can create, edit, delete, and manage public or private study groups, including meeting locations displayed using Google Maps.
(Fouzia)	Used Book Marketplace: Students can list used books with images, browse available books, and manage buying or selling activities. Book images are uploaded using Cloudinary.
(sayeefa)	Group Chat System: Members of a study group can communicate through group chat while tracking upcoming meetings from Google Calendar.
(tuli)	AI Team Recommendation: Based on users' skills, interests, and completed projects, the system recommends suitable teammates using Google Gemini.

Module 3	
Member	Feature Description
(Rayhan)	Group Join Request System: Students can send requests to join study groups, while group administrators can approve or reject requests.
(Rayhan)	Project Team Finder: Students can publish project recruitment posts, browse available teams, and view project meeting locations using Google Maps.
(Fouzia)	Tutor Finder: Students can search tutors by subject, availability, and ratings, and tutors can upload profile images and teaching materials using Cloudinary.
(Fouzia)	Resource Request System: Students can request for notes and study materials for a specific course and other tutors or students can upload the requested resources using Cloudinary storage.
(sayeefa)	Meeting Scheduler: Team leaders can create project meetings, schedule deadlines, and synchronize all events with Google Calendar.
(sayeefa)	File Sharing System: Team members can upload and organize project documents while associating files with scheduled meetings and deadlines.

(tuli)	Progress Dashboard: Students can monitor project progress, completed tasks, pending work, and team performance with AI-generated productivity insights.
(tuli)	Event & Workshop Announcement: Administrators can publish workshops, seminars, and hackathons, while Gemini recommends relevant events based on students' interests.